Physics 2: Physical Science & Technology (PHYC10004) SUM 2025

Week 3: Magnetism, Biot-Savart Law, Ampere's Law, Electromagnetic Induction, Maxwell's Equations and Magnetic Materials

Discussion Questions

1 A hydrogen atom that has lost its electron is moving east in the region where the magnetic field is directed from south to north. It will be deflected:

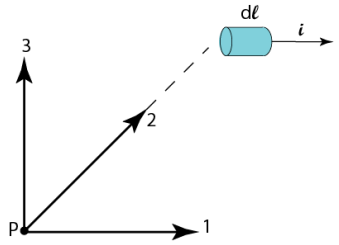
a) up b) down c) north d) south e) not at all.

2 Which of the following statements is correct:
When a charge moves in a uniform magnetic field (but not parallel to the magnetic field):

a) the magnitude of its velocity and momentum changes
b) the direction of its velocity and momentum changes
c) its kinetic energy changes
d) its acceleration changes.

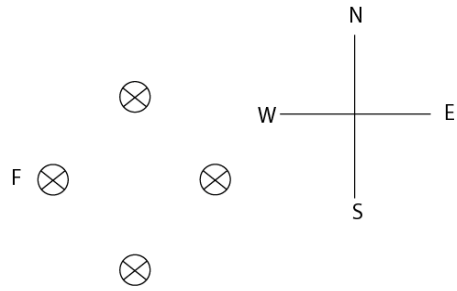
3 In the figure, the current element $i\mathbf{dl}$, the point P, and the three vectors (1, 2, 3) are all in the plane of page. The direction of the infinitesimal magnetic field $d\boldsymbol{B}$ generated by this current element at the point P is:

(a) in the direction marked "1"
(b) in the direction marked "2"
(c) in the direction marked "3"
(d) out of the page
(e) into the page.

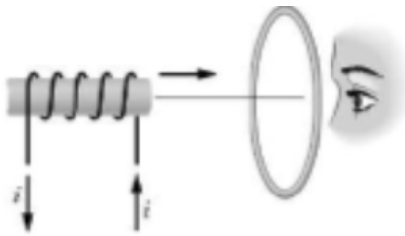


4 Four long straight wires carry equal currents into the page as shown. The magnetic force exerted on the wire F is:

(a) north (b) east (c) south (d) west (e) zero.



5 A current-carrying solenoid is moving towards a conducting loop as shown. In what direction does the current in the loop flow when viewed as shown?



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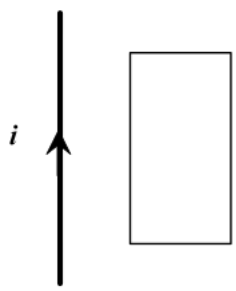

A long straight wire carrying a current *i* is in the plane of a rectangular conducting loop.

(a) What is the direction of the induced current in the rectangular loop if the current *i* is constant and being *moved toward* the rectangle?

(b) What is the direction of the induced current in the rectangular loop if the current *i* is *increasing* in the direction shown?

(c) What is the direction of the induced current in the rectangular loop if the current *i* is initially constant and then being shut off?

7 Suppose magnetic monopoles are discovered. What two major effects might this have on Maxwell's four equations of electromagnetism?

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Problem Solving Questions

8 An electron (mass $m = 9.1 \times 10^{-31}$ kg, charge $e = -1.6 \times 10^{-19}$ C) is projected with a velocity of 1.3×10^{7} m s^{-1} at an angle of 30° to a uniform magnetic field of strength $\boldsymbol{B} = 0.2$ T.

(a) Find the component of the velocity along the magnetic field initially.

(b) Find the component of the velocity perpendicular to the magnetic field initially.

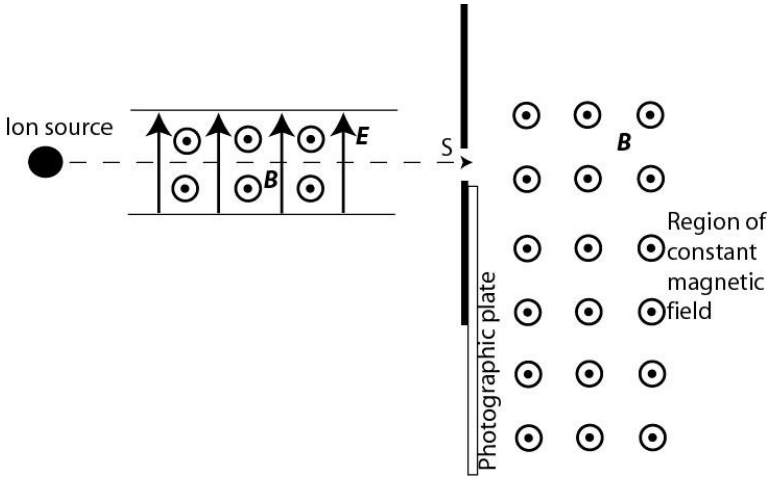
(c) Describe the trajectory of the electron.

(d) Find the radius of the resulting electron trajectory in the plane perpendicular to the magnetic field.

(e) Find the period of the motion.

(f) How far does the electron move along the field direction during one full revolution?

9 A mass spectrometer, sketched schematically below, has a source that produces *singly* charged ions. The ions are oxygen 16u (u is an atomic mass unit), carbon 12u, and an unknown element. The ions travel through perpendicular electric and magnetic fields, exit from a *narrow* slit S, and move in semicircular paths in a region of uniform magnetic field, as shown. Each species of ion makes a line on the photographic plate at the point where it strikes the plate. The carbon and oxygen lines are separated by 2.25 cm. The unknown element makes a line between them 1.16 cm from the carbon line.



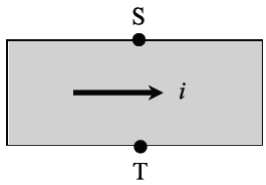
(a) Show that all ions entering the narrow slit S must have the same velocity. Hint: consider the relationship required between E and B.

(b) Which element gives rise to the line on the photographic plate furthest from the slit?

(c) Calculate the mass of the unknown element (in units of u).

10 *(The Hall Effect)*

A current runs from left to right in the rectangular conductor shown to the right below. The conductor is immersed in a magnetic field directed *into* the page. Point S is measured to be at a higher potential than point T.



(a) In what direction do charge carriers move inside the conductor if

- they are positively charged, and
- if they are negatively charged?

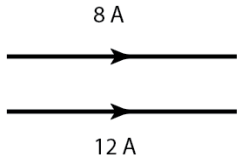
(b) What are the polarities of the upper and lower sides of the conductor?

(c) From (a), (b), and the direction of the magnetic field, deduce the sign of the charge carriers in the conductor.

11 A physicist is designing a cyclotron to accelerate protons to one-tenth the speed of light. The magnet used will produce a field of magnitude of 1.4 T. Ignoring relativistic effects, calculate (a) the required radius of the cyclotron 'Dees', and

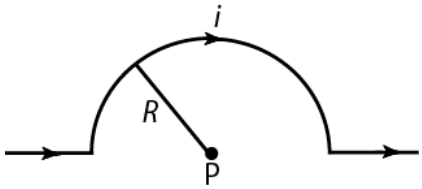
(b) the corresponding oscillator frequency.

12 Two long straight wires are parallel and carry current in the same direction. The currents are 8.0 and 12.0 A and the wires are separated by 0.40 cm.



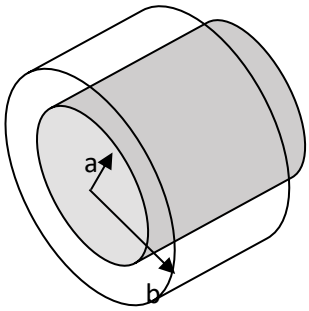
(a) Indicate the direction of the magnetic field at a point midway between the wires.
(b) Calculate its magnitude at the point midway between the wires.
(c) At what point between the wires is the magnetic field zero?

13 Using the Biot-Savart law show that the magnetic field at point P at the centre of the semicircle shown, is given by $B = \mu_0 I/4R$.



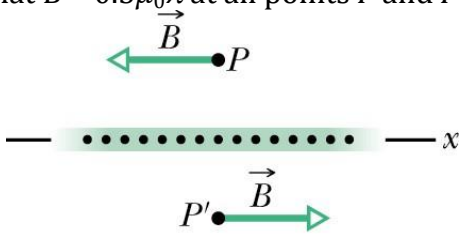
14
The windings of a very long coil (a solenoid) are made of a wire 0.8 mm in diameter. The turns just touch each other (but are made of insulated wire). Find the magnetic field inside it if it carries a current of 2 A.

15
Two thin, hollow, cylindrical, coaxial conductors with radii a and b ($b>a$) carry currents I which are equal in magnitude and oppositely directed. Ignoring the thickness of the conductors, demonstrate that the magnetic field in the volume:



(a) $r < a$, contained by the inner cylinder, is zero;
(b) $a < r < b$, between the conductors, is $B = \mu_0 I / 2\pi r$;
(c) $r > b$, outside the outer cylinder, is zero.

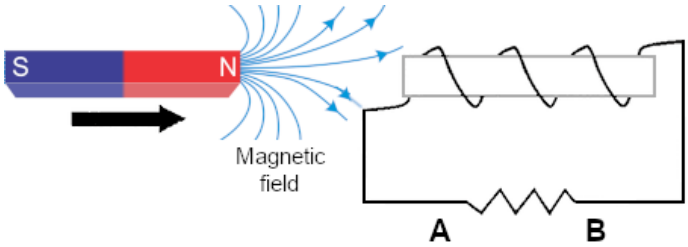
16 (mini-challenge question)
The figure below shows a cross section of an infinite conducting *sheet* carrying a current per unit x-length of λ; where the current emerges perpendicularly out of page.
(a) Use the Biot-Savart law and symmetry to show that for all points P above the sheet, and all points P' below it, the magnetic field **B** is parallel to the sheet and directed as shown.
(b) Use Ampere's law to prove that $B = 0.5\mu_0\lambda$ at all points P and P'.



17
A circular coil consisting of 200 turns and having a radius of 0.10 metre is placed perpendicular to a uniform magnetic field of induction 0.2 T. Find the *average* emf induced in the coil if, in 0.1 second:

(a) the field is doubled,
(b) the field is reduced to zero,
(c) the field is reversed in direction,
(d) the coil is rotated through 90 degrees,
(e) the coil is rotated through 180 degrees.

18 The coil shown in the figure has a radius of 50 mm and comprises 100 turns of wire. The resistor has a resistance of 100 ohms.

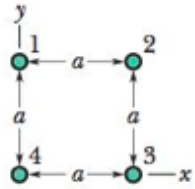


i) A bar magnet is moved towards the coil of wire as shown in the diagram. The coil has an air core. The magnet induces an emf in the coil that causes a current to flow. Explain the direction of the current flow through the resistor shown connected across the coil.

ii) Explain the direction of the current flow if the bar magnet is now moved away from the coil.

iii) With the bar magnet held fixed, the magnetic field is measured to be 0.4 T uniformly throughout the coil. The magnetic field is then reduced linearly to zero in 5 milliseconds. Determine the voltage that now appears across the resistor. Assume the resistance of the wire is negligible compared to the value of the resistor.

19 In the figure, four long straight wires are perpendicular to the page, and form a square pattern as shown, with an edge length of 13.5 cm. Each wire carries 7.5 A; the direction in wires 1 and 4 is out of the page; the direction for wires 2 and 3 is into the page. Calculate the net magnetic force per metre of length for wire 4. Express your answer in unit vector notation



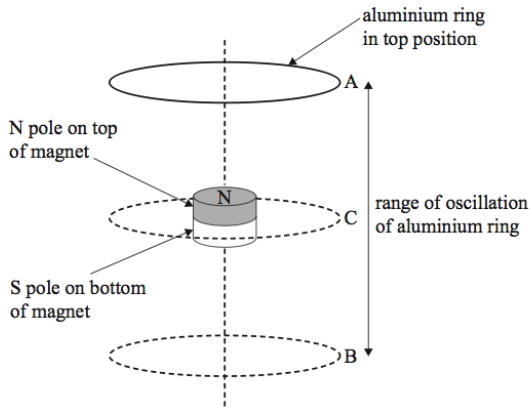
(reproduced from HRW 10th Edition, Chapter 29, Problem 37)

20 An aluminium ring oscillates vertically between point A and point B in the diagram below (point C is the midpoint of A and B). A strong, small bar magnet is *fixed* at point C.

(a) Sketch the variation of flux through the ring with time for one or two cycles. On your sketch show when the ring is at points A, B and C. Take $t = 0$ when the ring is at point A

(b) Sketch the shape of the variation of emf around the ring with time on the same set of axes

(c) When the ring is travelling *downwards* between points A and C, there is a current flowing around the ring. Identify the direction of the current flow (viewed from above), with reasons.



(Adapted from VCE Physics Examination 2013)

21

At what rate must the potential difference between the plates of a 2.0 μF parallel plate capacitor change to produce a displacement current i_D of 1.5 A? ($\varepsilon_0 = 8.85 \times 10^{-12}$ F/m)

(from HRW 10/e, Chapter 32 problem 13)

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Past Exam Questions

22

An alpha particle ($q = +2e, m = 6.65\times10^{-27}$kg) travels in a circular path of radius 4.50 cm in a uniform magnetic field with B = 1.20 T.

(i) Calculate its speed.

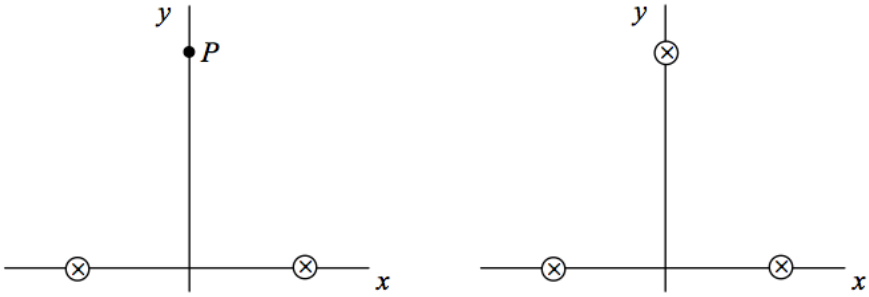
(ii) Calculate the frequency of revolution of the alpha particle.

23

Consider two very long parallel wires that are placed equal distances from the y-axis and carry equal currents directed into the page, as shown in the figure below left.

(a) What is the direction of the magnetic field at point P on the y-axis?

(b) If a long wire carrying a current into the page is now placed at point P (shown in the right figure) what will be the direction of the magnetic force on the wire? Explain your reasoning.

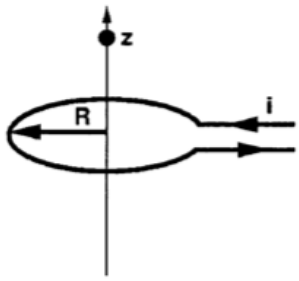


24

A circular loop of wire radius R carries a steady current i as shown in the diagram.

(i) Determine the magnitude of the magnetic field B_z at the point located a distance z m above the centre of the loop and along the axis of the loop. What is the direction of the z-component of $\vec{B}$ if the direction of i is as shown in the diagram?

(ii) Show that if $z \gg R$, the expression from part (i) simplifies to $B_Z \cong \frac{\mu_0 i R^2}{2z^3}$.



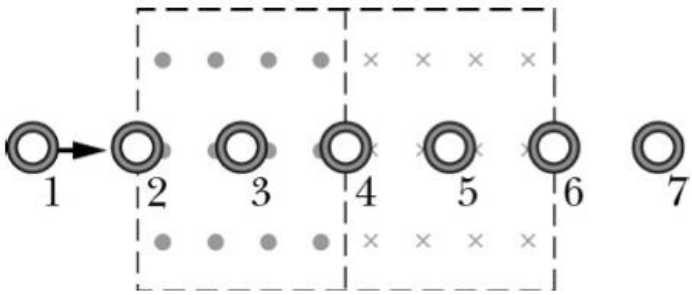
25 (2004)

Explain the difference between *paramagnetic*, *diamagnetic* and *ferromagnetic* materials and their behaviour in a non-uniform magnetic field. In your answer, you should include:

- A brief discussion of the magnetic nature of each type of material in terms of behaviour of the particles in the atom.
- Some discussion about the orientation of the magnetic dipole moments of atoms within a sample of each type of material in an external magnetic field.
- A description of the behaviour of a sample of these materials in a non-uniform magnetic field, clearly explaining (using diagrams) the role of the field's non-uniformity.

26

a) In the figure, a circular loop is moved at a constant velocity through regions where uniform magnetic fields of the same magnitude are directed into or out of the page. The field is zero everywhere else.



At which one or more of the seven indicated loop positions is the current induced in the loop:

i) Clockwise;

ii) Anti-clockwise;

iii) Zero?

b) Two diamagnetic spheres (1 & 2) are located near the south pole of a bar magnet.

i) Are the magnetic dipole moments of the spheres directed toward or away from the bar magnet?
ii) Are the magnetic forces on the spheres directed toward or away from the bar magnet? Explain using diagrams if necessary.
iii) Is the magnetic force on sphere 1 greater than, less than or equal to that on sphere 2?

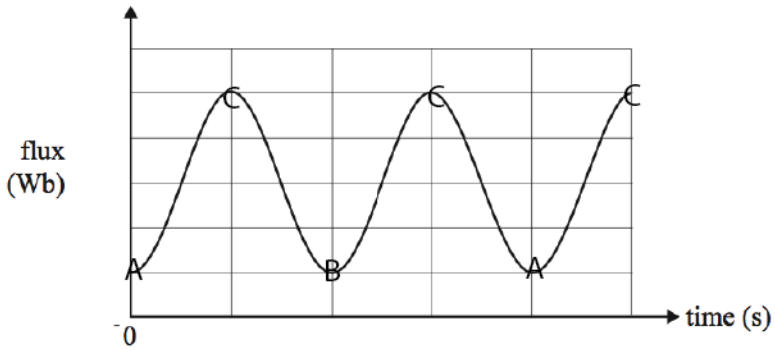
Short solutions:

Discussion Questions:

1a 2b&d 3e 4b 5 clockwise **6a** anticlockwise **6b** anticlockwise **6c** clockwise
7 This was covered in the lecture notes on the LMS – Gauss' Law for magnetism would now be non-zero, and there would be an extra term in Faraday's Law where a moving magnetic charge would produce an electric field.

Problem Solving Questions:

8 a) 1.1×10^{7}m/s **b)** 6.5×10^{6}m/s **c)** helical **d)** 1.85×10^{-4}m **e)** 1.79×10^{-10}s **f)** 2×10^{-3}m
9 a) v=E/B **b)** oxygen **c)** 14u **10a)** +ve-left-right, -ve-right-left **b)** S=+ve, T=-ve **c)** +ve
11 a) 0.224m **b)** 2.13×10^{7}Hz **12 a)** out of page **b)** 4.0×10^{-4} T **c)** 0.16 cm from 8V wire
14) 3.14×10^{-3} T
17 a 12.6 V **b** 12.6 V**c** 25.1 V**d** 12.6 V **e** 25.1 V**18i** A to B in the resistor **18ii** B to A in the resistor
18iii 62.8 V **19** (-0.125**i**, +0.0417**j**) units are mN/m
20a



20b Same shape as answer to 5a, but displaced one half horizontal division L or R (depending on the direction taken for positive current), and averaging zero. The emf is zero when the ring is at points A, B and C. **20c** current flows clockwise
21 7.5×10^{5} V/s

Past Exam Questions:

22 i 2.6×10^{6} m/s **ii** 9.2 MHz **23 a** positive x **b** negative y

24i $B=\dfrac{m_0 i 2pR^2}{4p(z^2+R^2)^{\frac{3}{2}}}$

25 *Suggest that you write you answer in dot points. Each dot should address one of the aspects of the question. Use diagrams wherever possible. You will find good summaries in the text (HRW; sections 32.5 through to 32.8) and also Wikipedia and Hyperphysics.*
26a) At 1: zero; 2: CW; 3: zero; 4: ACW; 5: zero; 6: CW; 7: zero **b)** away; away; less than